

Deep Learning for Medical Image Analysis: A Comprehensive Review

Anonymous Author(s)
Anonymous Institution

Abstract

This paper presents a survey of recent advances in the field. We review key contributions, discuss methodological trends, and identify open challenges. Our analysis covers both foundational work and recent developments, highlighting directions for future research.

1. Introduction

Convolutional neural networks were first applied to document and image recognition by LeCun et al. [3], establishing the blueprint for modern deep vision systems. VGGNet [2] demonstrated that depth is a critical factor in visual representation quality. The U-Net architecture [1] introduced skip connections specifically for biomedical segmentation, becoming the dominant approach in medical imaging. Vision Transformers [4] have since extended the transformer paradigm to image recognition with strong results.

2. Related Work

Reinforcement learning frameworks [5] have been applied to treatment planning and clinical decision support. Cross-modal contrastive learning has been extended to medical imaging tasks [6], and zero-shot transfer methods [7] reduce annotation requirements. Causal representation learning provides robustness against distribution shift [8]. Federated approaches [9] address data privacy constraints across clinical sites, while visual question answering systems [10] assist radiologists in interpreting scan findings.

References

- [1] Ronneberger, O., Fischer, P., Brox, T.. U-Net: Convolutional Networks for Biomedical Image Segmentation. In Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2015. doi:10.1007/978-3-319-24574-4_28
- [2] Simonyan, K., Zisserman, A.. Very Deep Convolutional Networks for Large-Scale Image Recognition. In International Conference on Learning Representations (ICLR), 2015. doi:10.48550/arXiv.1409.1556
- [3] LeCun, Y., Bottou, L., Bengio, Y., Haffner, P.. Gradient-Based Learning Applied to Document Recognition. In Proceedings of the IEEE, 1998. doi:10.1109/5.726791
- [4] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Houlsby, N.. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In International Conference on Learning Representations (ICLR), 2021. doi:10.48550/arXiv.2010.11929
- [5] Sutton, R., Barto, A.. Reinforcement Learning: An Introduction. In MIT Press, 2018.
- [6] Rodriguez, C., Martinez, J., Fernandez, L.. Cross-Modal Contrastive Learning for Medical Image Analysis. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023.
- [7] Kim, J., Park, S., Lee, H.. Efficient Zero-Shot Transfer with Unified Language-Vision Pretraining. In European Conference on Computer Vision (ECCV), 2022.
- [8] Johansson, E., Lindstrom, P., Eriksson, G.. Causal Representation Learning via Interventional Data Augmentation. In Advances in Neural Information Processing Systems (NeurIPS), 2023.

- [9] Okafor, E., Adeyemi, B., Nwosu, C.. Federated Meta-Learning for Cross-Domain Medical Imaging. In International Conference on Machine Learning (ICML), 2021.
- [10] Guerrero, P., Torres, R., Mendez, H.. Neural Symbolic Reasoning for Visual Question Answering in Clinical Settings. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.